

The HBond Calculator

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1 What the calculator computes

The HBond calculator describes a through-space tensor associated with backbone hydrogen bonding. It starts from hydrogen-bond partners reported by DSSP, resolves each partner to a donor backbone N atom and an acceptor backbone O atom, and treats the $N \cdots O$ line as an oriented source. For every atom in the conformation, each accepted source contributes a 3×3 geometric tensor with units $\text{\AA}^{-3}$. The observable in the background is the nuclear shielding tensor, and after referencing it is the chemical shift, but this calculator does not output ppm shieldings. It outputs the geometric part that can correlate with a DFT shielding contribution after calibration.

The physical idea is that a hydrogen bond can reorganize electron density around the donor and acceptor groups. In an applied magnetic field, that electronic response can change the secondary magnetic field felt at a nearby nucleus. The code does not compute the electronic response. It keeps geometry used as a proxy for part of the angular signal: the donor-to-acceptor direction, the source midpoint, and the vector from that midpoint to the atom being evaluated.

2 Where shielding enters

For a nucleus in an applied magnetic field B_0 , the local field is written

$$B_{\text{nuc}} = (I - \sigma)B_0. \quad (1)$$

Here I is the 3×3 identity matrix and σ is the shielding tensor. The tensor is dimensionless in physical NMR: it is the linear coefficient connecting a small applied field to the electronic field that opposes or augments it at the nucleus. A matrix is needed because a protein conformation is not spherical. A field applied along z , for example, can induce a secondary field whose x , y , and z components depend on how the local groups are oriented.

Equation (1) is a quantum response statement. The applied field perturbs the electronic state, the perturbed electrons produce their own magnetic field, and σ records the linear part of that response. The HBond calculator replaces that electronic calculation by an oriented geometric source. It says: if DSSP reports that one peptide $N - H$ donates to another peptide $C = O$, then the line from the donor N to the acceptor O is a local axis that may organize the shielding response. The tensor form keeps that axis as an angular object rather than reducing it to a distance count.

The solution NMR scalar most often compared across atoms is the isotropic chemical shift, related to $\text{tr } \sigma / 3$ after reference subtraction and sign conventions. This calculator keeps the full rank-2 descriptor because the anisotropic part records how the signal changes with molecular orientation. That angular part is the content used when the descriptor is compared with tensor-valued DFT shielding data.

3 The tensor split used by the code

The code stores every 3×3 tensor X through the `SphericalTensor` decomposition. This projects the matrix into three pieces that rotate independently: a scalar, an axial vector, and a symmetric traceless tensor. It is not an eigendecomposition. The scalar part is

$$T_0 = \frac{1}{3} \text{tr } X, \quad (2)$$

the average of the three diagonal entries. It is the isotropic component.

The antisymmetric part is stored as the three-component vector

$$\mathbf{T}_1 = \frac{1}{2} \begin{pmatrix} X_{yz} - X_{zy} \\ X_{zx} - X_{xz} \\ X_{xy} - X_{yx} \end{pmatrix}. \quad (3)$$

This vector is the compact dual of the matrix part that changes sign under transpose. It is usually not observed in standard isotropic solution NMR, but the HBond tensor can contain it because its main term is generally asymmetric.

The remaining part is the symmetric traceless matrix

$$\mathbf{S} = \frac{\mathbf{X} + \mathbf{X}^\top}{2} - T_0 \mathbf{I}, \quad \text{tr } \mathbf{S} = 0. \quad (4)$$

It has five independent entries. The code writes them in the real spherical T_2 basis

$$\mathbf{T}_2 = \begin{pmatrix} \sqrt{2} S_{xy} \\ \sqrt{2} S_{yz} \\ \sqrt{3/2} S_{zz} \\ \sqrt{2} S_{xz} \\ (S_{xx} - S_{yy})/\sqrt{2} \end{pmatrix}. \quad (5)$$

The factors in Eq. (5) are part of the file format. They preserve the Frobenius norm: the squared length of this five-vector equals $\sum_{ab} S_{ab}^2$. HBond populates all three channels, T_0 , T_1 , and T_2 , by decomposing the accumulated full tensor for each atom.

4 The method implemented

Hydrogen-bond sources come from `DsspResult`. Here, `DsspResult` calls `libdssp`, which implements the DSSP Kabsch–Sander hydrogen-bond assignment. `HBondResult` does not recompute that energy, inspect the stored energy values, or apply a DSSP energy threshold. It uses every partner residue index that `DsspResult` reports, then applies its own geometric filters.

For each residue i , the code reads up to two acceptor partners and two donor partners. On the donor side, residue i 's backbone N donates to a partner residue's backbone O . On the acceptor side, a partner residue's backbone N donates to residue i 's backbone O . A candidate source is discarded if either atom is missing, if the donor and acceptor residues are less than 2 residues apart in sequence, if the pair has already been seen, if the $N \cdots O$ distance is below 0.1 Å, or if that distance is above 50.0 Å. The last two values are `singularity_guard_distance` and `hbond_dipolar_max_distance` in `data/calculator_params.toml`. No local donor–hydrogen–acceptor angle test appears in this calculator.

An exact way to hold the model is to replace each accepted hydrogen bond by a short oriented segment from donor N to acceptor O , then keep its midpoint and direction. Those are the objects the code stores. If the donor and acceptor positions are $\mathbf{n}_k$ and $\mathbf{o}_k$, then

$$\mathbf{m}_k = \frac{\mathbf{n}_k + \mathbf{o}_k}{2}, \quad L_k = |\mathbf{o}_k - \mathbf{n}_k|, \quad \hat{\mathbf{h}}_k = \frac{\mathbf{o}_k - \mathbf{n}_k}{L_k}. \quad (6)$$

$\mathbf{m}_k$ is the source point in Å, L_k is the $N \cdots O$ length in Å, and $\hat{\mathbf{h}}_k$ is a unitless direction from donor to acceptor.

For a target atom at $\mathbf{r}_i$, the code forms

$$\mathbf{d}_{ik} = \mathbf{r}_i - \mathbf{m}_k, \quad r_{ik} = |\mathbf{d}_{ik}|, \quad \hat{\mathbf{d}}_{ik} = \frac{\mathbf{d}_{ik}}{r_{ik}}, \quad c_{ik} = \hat{\mathbf{d}}_{ik} \cdot \hat{\mathbf{h}}_k. \quad (7)$$

Here r_{ik} is the atom-to-midpoint distance in Å, and c_{ik} is the cosine of the angle between the source direction and the midpoint-to-atom direction. The dyadic product $u_a v_b$ is a 3×3 table: it takes a component along v and returns a vector along u . The HBond contribution is

$$M_{ab}^{(ik)} = \frac{9c_{ik} \hat{d}_{ik,a} \hat{h}_{k,b} - 3\hat{h}_{k,a} \hat{h}_{k,b} - (3\hat{d}_{ik,a} \hat{d}_{ik,b} - \delta_{ab})}{r_{ik}^3}. \quad (8)$$

The Cartesian indices a, b run over x, y, z , and δ_{ab} is one when $a = b$ and zero otherwise. The numerator is dimensionless, so $M^{(ik)}$ has units $\text{\AA}^{-3}$. The first term uses both the viewing direction $\hat{d}$ and the hydrogen-bond direction $\hat{h}$, which is why the tensor can be asymmetric. The last term is a traceless line-of-sight tensor. Taking the trace gives

$$\frac{1}{3} \text{tr } M^{(ik)} = f_{ik} = \frac{3c_{ik}^2 - 1}{r_{ik}^3}. \quad (9)$$

The scalar f_{ik} is accumulated as a separate feature over the same accepted atom–source pairs.

Before adding Eq. (8), the code applies three atom–source filters. First, r_{ik} is accepted at or above 0.1 Å, preventing the r^{-3} expression from being evaluated at a singular point. Second, the target atom may not be the donor N or acceptor O of that source. Third, the target is accepted outside the near field of the finite $N \cdots O$ segment:

$$r_{ik} > \alpha L_k, \quad \alpha = 0.5. \quad (10)$$

The parameter α is `near_field_exclusion_ratio`. The point source description is not used inside half the segment length. The calculator does not use a spatial search cutoff for atom–source evaluation; after DSSP has supplied the source list, all sources are considered and the r^{-3} decay sets their size.

For each atom, the stored tensor is

$$M_i = \sum_{k \in \mathcal{A}_i} M^{(ik)}, \quad (11)$$

where $\mathcal{A}_i$ is the set of resolved hydrogen bonds that pass the filters for atom i . The nearest accepted source is tracked by r_{ik} . The code also counts accepted sources with $r_{ik} < 3.5 \text{ \AA}$, the configured `hbond_counting_radius`. The nearest-distance and count features are geometric diagnostics; they do not change the tensor sum. The `SampleKernelAt` path evaluates the same sum at an arbitrary point using the stored source list. Because that point has no atom index, it applies the singularity and near-field checks but not the donor/acceptor endpoint exclusion.

The main approximations are visible in these equations. Each hydrogen bond is reduced to one midpoint and one $N \rightarrow O$ axis. All accepted sources have unit weight; the code does not multiply by DSSP energy, hydrogen-bond length dependence, partial charge, susceptibility, or an empirical ppm coefficient. It uses backbone DSSP partners, so side-chain and water hydrogen bonds are outside this calculator. It uses $N \rightarrow O$, not $H \rightarrow O$, as the tensor axis after DSSP has identified the donor and acceptor.

5 External inputs and output

HBondResult consumes atom positions and residue topology from the current protein conformation, plus DSSP partner data from `DsspResult`. DSSP is the external library boundary: `DsspResult` calls `libdssp`, then maps its secondary-structure and backbone hydrogen-bond partner slots back onto the conformation. `SpatialIndexResult` is declared as a dependency, but this calculator iterates over the resolved hydrogen bonds directly rather than querying a neighbor list.

MOPAC charges and bond orders, AIMNet2 charges or embeddings, ff14SB partial charges, and APBS electric fields are not inputs to this calculator. They feed other environment and electrostatic descriptors in the pipeline. Here the written files are `hbond_shielding.npy`, an $N \times 9$ array in the $[T_0, T_1, T_2]$ layout of Eqs. (2)–(5), and `hbond_scalars.npy`, an $N \times 4$ array containing nearest atom-to-midpoint distance, its $1/r^3$, the count inside 3.5 Å, and $\sum f_{ik}$. If no source passes the filters for an atom, these scalar columns

and the tensor remain at their zero initial values. Despite the file name, the tensor array is a geometric quantity in $\text{\AA}^{-3}$. Calibration against DFT reference tensors is what maps it to a shielding contribution.